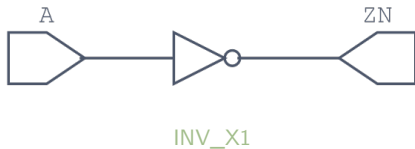
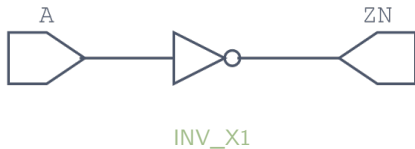


Liberty: units and characterization conditions



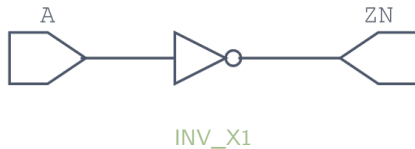
```
/* NangateOpenCellLibrary_typical.lib */  
library (NangateOpenCellLibrary) {  
    delay_model : table_lookup;  
    time_unit : "1ns";  
    leakage_power_unit : "1nW";  
    voltage_unit : "1V";  
    current_unit : "1mA";  
    capacitive_load_unit (1,ff);  
    nom_process : 1.00;  
    nom_temperature : 25.00;  
    nom_voltage : 1.10;  
    /* Other groups omitted */  
}
```

Liberty: power pins and signal-pin supplies



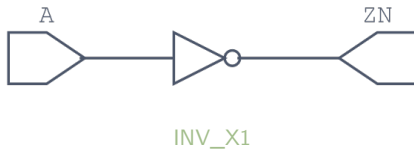
```
/* NangateOpenCellLibrary_typical.lib */
/* Within cell (INV_X1) */
pg_pin(VDD) {
    voltage_name : VDD;
    pg_type      : primary_power; ← supply identity
}
pg_pin(VSS) {
    voltage_name : VSS;
    pg_type      : primary_ground;
}
/* Within signal pins A and ZN */
related_power_pin : "VDD"; ← signal-pin association
related_ground_pin : "VSS";
```

Liberty: input capacitance and logic function



```
/* NangateOpenCellLibrary_typical.lib */
cell (INV_X1) {
  area : 0.532000;
  pin (A) {
    direction : input;
    capacitance : 1.700230;
  }
  pin (ZN) {
    direction : output;
    function : "!A"; ← Boolean function
    max_capacitance : 60.730000;
    /* Timing/power groups omitted */
  }
  /* Other attributes omitted */
}
```

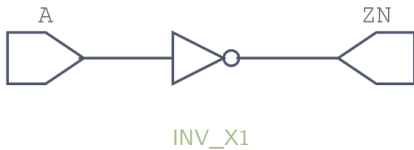
Liberty: define the measurement thresholds



```
/* NangateOpenCellLibrary_typical.lib */
input_threshold_pct_rise : 50.00; ← delay: 50% → 50%
input_threshold_pct_fall : 50.00;
output_threshold_pct_rise : 50.00;
output_threshold_pct_fall : 50.00;

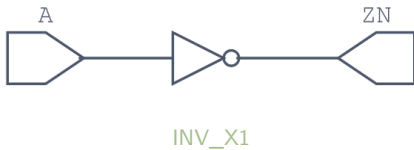
slew_lower_threshold_pct_rise : 30.00; ← slew: 30% → 70%
slew_upper_threshold_pct_rise : 70.00;
slew_lower_threshold_pct_fall : 30.00;
slew_upper_threshold_pct_fall : 70.00;
```

Liberty: a timing arc from A to ZN



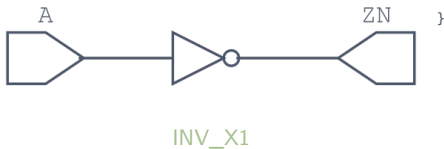
```
/* NangateOpenCellLibrary_typical.lib */
/* Within INV_X1 / pin (ZN) */
timing () {
  related_pin : "A"; ← input-to-output arc
  timing_sense : negative_unate;
  cell_fall (Timing_7_7) {
    /* 7 x 7 delay table */
  }
  cell_rise (Timing_7_7) {
    /* 7 x 7 delay table */
  }
  fall_transition (Timing_7_7) {
    /* 7 x 7 output slew table */
  }
  rise_transition (Timing_7_7) {
    /* 7 x 7 output slew table */
  }
}
```

Liberty: index delay by slew and load

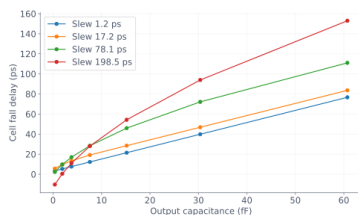


```
/* NangateOpenCellLibrary_typical.lib */  
lu_table_template (Timing_7_7) {  
    variable_1 : input_net_transition;  
    variable_2 : total_output_net_capacitance;  
    index_1 ("0.0010,0.0020,0.0030,0.0040,0.0050,0.0060,\n              0.0070");  
    index_2 ("0.0010,0.0020,0.0030,0.0040,0.0050,0.0060,\n              0.0070");  
}
```

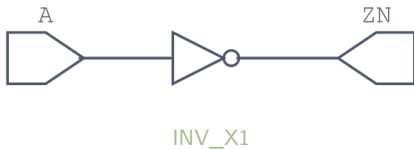
Liberty: the delay table in numbers



```
/* NangateOpenCellLibrary_typical.lib */
cell_fall (Timing_7_7) {
  index_1 ("0.00117378,0.00472397,0.0171859,\
           0.0409838,0.0780596,0.130081,0.198535");
  index_2 ("0.365616,1.897810,3.795620,7.591250,\
           15.182500,30.365000,60.730000");
  /* First row; other six rows omitted */
  values ("0.00334769,0.00529785,0.00763425,\
           0.0122592,0.0214710,0.0398747,0.0766650");
}
```

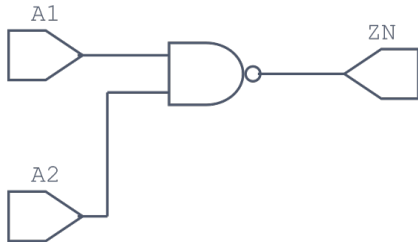


Liberty: leakage and internal switching power



```
/* NangateOpenCellLibrary_typical.lib */  
/* Within INV_X1 */  
cell_leakage_power : 14.353185;  
leakage_power () {  
    when : "!A";  
    value : 10.102224;  
}  
leakage_power () {  
    when : "A";  
    value : 18.604146;  
}  
/* Within pin (ZN) */  
internal_power () {  
    related_pin : "A";  
    fall_power (Power_7_7) { /* omitted */ }  
    rise_power (Power_7_7) { /* omitted */ }  
}
```

NAND2: two inputs, two timing arcs



NAND2_X1

```
/* NangateOpenCellLibrary_typical.lib */
/* Within cell (NAND2_X1) */
pin (A1) {
    capacitance : 1.599032;
    /* Other attributes omitted */
}
pin (A2) {
    capacitance : 1.664199;
    /* Other attributes omitted */
}
pin (ZN) {
    function : "!(A1 & A2)";
    timing () {
        related_pin : "A1"; ← one timing group per input
        timing_sense : negative_unate;
        /* Tables omitted */
    }
    timing () {
        related_pin : "A2";
        timing_sense : negative_unate;
        /* Tables omitted */
    }
}
```